Cooperative Task Allocation and Path Planning for Multi-UAVs in Low-Altitude Urban Intelligent Transportation Systems

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***Abstract*—In low-altitude urban intelligent transportation systems, efficient cooperative task allocation and path planning for multiple unmanned aerial vehicles (UAV) are critical for ensuring the effective execution of complex tasks. This paper proposes a distributed decision-making and autonomous planning framework to achieve cooperative task allocation and path planning for multi-UAVs in low-altitude urban traffic environment. The mission requirements of task allocation and path planning are modeled using evolutionary potential games and show that there exists a Nash equilibrium for the proposed potential function. An Improved Log-linear Learning Algorithm (ILLA) is proposed, and suitable Boltzmann parameters are derived which will enable the proposed ILLA to converge to the optimal Nash equilibrium with a probability one. Furthermore, a Constraint-Based Multi-layer Bidirectional Adaptive A-Star (CBMBA A-Star) algorithm is designed to find optimal and collision free paths for each UAV. Compared with the baseline method, simulation results demonstrate that the proposed approach improves the task reward by 11.67%, reduces the task execution time by 37.41%, and decreases run time by 61.02%, confirming its effectiveness and efficiency in the complex low-altitude urban traffic scenario.**

***Index Terms*—Unmanned aerial vehicle (UAV), low-altitude economy, task allocation, path planning, game theory.**

I. Introduction

LOW-ALTITUDE aerial vehicles are the main carriers of the low-altitude economy. With their unique flight capabilities and wide application scenarios, they have promoted the application and development of multiple traffic scenarios such as logistics transportation, emergency rescue, autonomous driving, and urban air mobility [1], [2], [3], [4]. With the in-depth integration of low-altitude aerial vehicle technology and low-altitude urban intelligent transportation systems (LU-ITS), new solutions for low-altitude traffic have been provided, and a more promising blueprint for the development of the future low-altitude economy has been depicted. Existing unmanned systems are mainly designed and developed for a single UAV, with simple tasks, single scenarios, and low safety. They rarely consider the autonomous planning and scheduling management among swarm vehicles, making it difficult to meet the safe and efficient flight requirements of large-scale aerial vehicles in LU-ITS [5], [6].

To improve the flight efficiency of aerial vehicles, reduce operational costs, and adapt to complex low-altitude urban traffic scenarios and multi-task scheduling requirements, autonomous aerial vehicles can be adopted. Task allocation and path planning are the core and crucial link to realizing the autonomous, efficient and collaborative operation of aerial vehicles and ensuring the accurate execution of tasks. Besides analyzing mission rewards, costs, and constraints in LU-ITS that target emergency rescue and last-mile cargo delivery tasks, precise modeling of collaborative relationships and obstacle avoidance behaviors among multi-UAVs is essential. During the task allocation and path planning processes in LU-ITS, UAVs rely on communication networks to continuously interact with neighboring UAVs and update their state information, which is closely associated with mission-related rewards and costs. Then, any two UAVs are subject to physical constraints [7], which encompass minimum safe separation distance, turning angle limits, and velocity bounds. They continuously exchange data and update states iteratively until a globally optimal planning solution is obtained. These collaborative interactions impose strict spatiotemporal requirements on UAVs. However, effective methods that can realize real-time and efficient task allocation and path planning, while being tailored to diverse mission requirements in LU-ITS, are still lacking.

To fill this gap, a decision-making and planning method integrating game theory and graph search is proposed to achieve cooperative task allocation and path planning of multi-UAVs in the application of traffic emergency rescue and cargo

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transportation in low altitude urban environment. The main contributions are summarized as follows.

1) A game-theoretic based distributed and autonomous planning framework is proposed in the low-altitude urban traffic environment, which facilitates information exchange among UAVs through iterative strategy games, enabling the rapid attainment of a globally optimal and stable solution for task allocation and path planning.

2) The mathematical form of the optimal potential function is analytically derived, and it is demonstrated that the proposed Improved Log-linear Learning Algorithm (ILLA) can converge to the optimal Nash equilibrium with probability one by adopting appropriate Boltzmann parameters.

3) The constraint-based multi-layer bidirectional adaptive A-Star algorithm (CBMBA A-Star) is proposed, which is a graph search-based approach capable of generating optimal and collision-free paths in low altitude urban environment with bandit threats.

4) The proposed method is integrated into multi-UAVs collaborative task allocation and path planning, and its optimality, scalability, and robustness have been verified through simulation experiments in urban traffic emergency rescue and cargo transportation scenarios.

The rest of this paper is organized as follows. Section II reviews the existing literature on the applications of task allocation and path planning for unmanned autonomous vehicles in intelligent transportation systems and low altitude environments. In Section III, we model the problem of cooperative task allocation and path planning for multi-UAVs in LU-TIS and propose an autonomous decision-making framework based on game theory. Section IV describes the our optimization algorithm for seeking Nash equilibrium and collision-free paths, and the convergence and optimality of the algorithm is proved. Numerical simulations in Section V confirm that the proposed approaches outperform existing technologies. Finally, Section VI concludes this paper.

II. Related Work

The core of cooperative task allocation and path planning in low-altitude urban intelligent transportation systems essentially is a combinatorial optimization problem [8], [9]. Particularly in LU-ITS scenarios, the traffic emergency response, last-mile delivery and uncertain constraints must be considered. Existing research is primarily categorized into four branches: Classical Optimization Methods, Heuristic Search Methods, Game Theory Methods, and Machine Learning Methods.

A. Classical Optimization Methods for Task Allocation and Path Planning

Cooperative task allocation and path planning based on classical optimization methods involves constructing mathematical models to transform a series of issues. Deterministic or numerical programming methods are used to achieve the optimal or near-optimal performance in LU-ITS. To tackle ITS-oriented task allocation challenges from complex urban terrain and strong task coupling, Yao et al. [10] proposes an evolutionary utility prediction matrix method to develop a distributed framework for heterogeneous task planning. By analyzing input-output coupling in typical urban traffic-related task modes, they optimize each sub-module design, thereby improving the accuracy and efficiency of cooperative task allocation. A low-altitude economy network framework, formulating a long-term energy optimization problem for trajectory generation and resource allocation is proposed [11]. Then, a novel diffusion-based trajectory generation algorithm is further developed to generate optimal trajectories with guided sampling. Srinivasan et al. [12] prensents a branch-and-bound-based planning algorithm to realize multi-objective coverage task allocation and path planning in search and rescue operations. However, these classical optimization methods require accurate modeling of tasks, constraints, and system states. Particularly in low-altitude urban scenarios involving large-scale tasks or multiple constraints, solving the model is time-consuming, making them unsuitable for task planning scenarios with strict real-time requirements.

B. Heuristic Search Methods for Task Allocation and Path Planning

Heuristic search methods are widely applied in cooperative task allocation and path planning of low-altitude autonomous vehicles in LU-ITS due to their simplicity in operation, strong robustness, and capability to obtain approximate optimal solutions within a limited time [13], [14]. To address efficient flight routing and scheduling in air traffic flow management, a discrete-time flow dynamic model and a bottom-up hierarchical approach are proposed for individual flight plans [15]. These methods guide the population to converge to the feasible region through an adaptive penalty mechanism, balancing the diversity and convergence of solutions. To meet the critical demand for rapid, accurate optimal path planning in IoT-enabled UAV delivery systems, a multi-strategy particle swarm optimization (PSO) algorithm is presented, which integrates local deadlock jump, adaptive nonlinear inertia weights, and multi-population differentiated evolution [16]. Based on the consensus-based bundle algorithm (CBBA), Wang et al. [17] optimize the start times and consensus phases to effectively mitigate task sequence conflicts, which achieves the decentralized task allocation of multi-UAVs. Zhang et al. [18] proposed an enhanced A-Star algorithm for UAV path planning in complex 3D environments. Numerical simulations show that the algorithm is more efficient regarding computation, path cost and safety. Chai et al. [19] proposed a high quality path planner based on the multi-strategy fusion differential evolution (MSFDE) algorithm. The method integrates three new strategies to balance development and exploration capabilities. For UAV dynamic path planning in low-altitude complex urban environments, a bi-level offline-online algorithm is proposed [20], where an improved hunger games search (HGS) generates optimized paths with static obstacle constraints, and a improved rapid-exploring random tree (RRT) algorithm is adopted to achieve online path planning. While performing well in simulations, these methods lack a unified evaluation system in practical low-altitude air traffic scenarios and have weak model interpretability. Their adaptability in operational environments with communication and strict behavioral constraints also needs further verification.

C. Game-Theoretic Methods for Task Allocation and Path Planning

In recent years, game theory, as an important theoretical tool for studying interactions and strategy selection among multiple aerial entities has shown promising application prospects in multi-entity cooperative tasks within intelligent transportation systems [21], [22]. Each autonomous vehicle aims to maximize its own payoff and gradually converges to a Nash equilibrium through strategic games, thereby achieving cooperative optimization at the system level. A spatial game model is developed for incorporating parking operators as game players into the analytical framework and accounting for heterogeneity in travelers' income [23], to evaluate the impact of various urban transportation policies on the competition among travel modes for commuters on the corridor. Chen et al. [24] proposes a distributed online task offloading and resource allocation scheme based on high-altitude platforms and UAVs to minimize ground device energy consumption. The problem is decomposed into two sub-problems via stochastic optimization with a distributed algorithm designed using game theory. Yazicioglu et al. [25] proposed a distributed planning method for multi-vehicles system to provide optimal services, and a best response leanrning algorithm (BRLA) is presented to converge to Nash equilibrium and ensure finite sub-optimality. Addtionally, log-linear learning algorithm (LLA) offers an effective way to seek the Nash equilibrium of the games [26], [27]. Li and Duan [28] proposed cooperative search and surveillance method based on potential game, and use a binary log-linear learning method to realize the motion control of UAVs in the complex urban environment. A three-level decision-making framework based on normal-form game is proposed [29]. It designs a payoff function balancing safety rewards and traffic rule compliance, enabling intelligent decisions in emergencies that account for surrounding vehicles' behavior while balancing safety and rules. However, with the increasing complexity of low-altitude urban air traffic scenarios, factors such as the tight coupling between tasks and routes, communication and response time constraints, and dynamic threats have restricted the widespread application of game theory in large-scale UAVs cooperative task allocation and path planning.

D. Machine Learning Methods for Task Allocation and Path Planning

Recently, for cooperative task allocation and path planning based on machine learning methods in air intelligent transportation systems, aerial entities can make decisions without relying on accurate modeling by learning interactions with the aerial environment or other aerial entities [1], [5], [30]. An air traffic density prediction method based on historical data and machine learning is proposed to assist air traffic controlauthorities in planning resources in advance and evaluating sector complexity through structuring traffic flow and predicting its distribution [31]. For air intelligent transportation systems, a

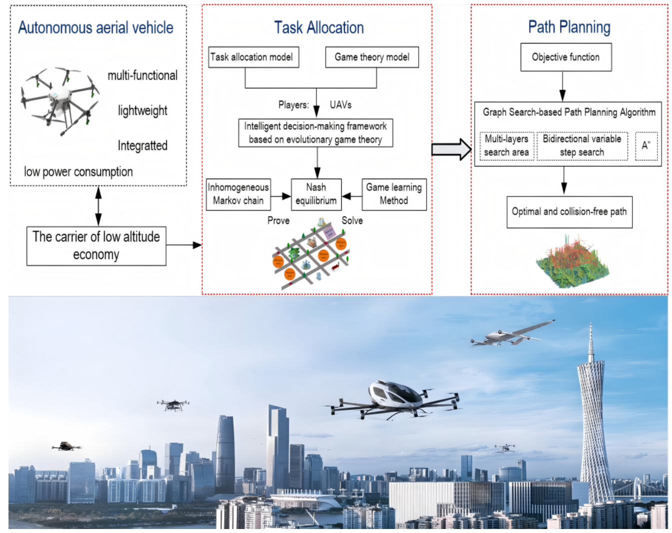

Fig. 1. Framework of cooperative mission planning for multi-UAVs.

hierarchical group-aware graph neural network model is proposed to capture latent dependencies between spatially distant but highly correlated airspace partitions via airspace partition graphs, regional correlation graphs and a differentiable grouping network, with experiments proving it outperforms existing models on real datasets [32]. Seid et al. [33] employs multi-agents deep reinforcement learning to realize computational offloading and resource allocation for multiple UAVs in the Internet of Things (IoT), optimizing computational costs and service quality. Kuma et al. [34] presents a deep reinforcement learning-based joint optimization method for grasping and placing, which exhibits strong adaptability in dynamic and unstructured environments. A novel framework integrating strategic-level demand-capacity balance and tactical-level reinforcement learning for safety separation management in urban air mobility [35]. Experiments demonstrate that the hybrid method ensures safety standards while achieving higher operational efficiency than standalone reinforcement learning or alternative schemes. Deep reinforcement learning-based cooperative task planning is suitable for scenarios with high uncertainty and incomplete information, featuring excellent generalization ability. However, its reliance on large amounts of interaction data leads to high training costs, which limits its deployment efficiency in practical complex environments.

III. Problem Formulation and System Model

A. Problem Formulation

We focus on multi-UAVs collaborative task allocation and path planning in low-altitude urban traffic scenarios, a critical component of missions such as air emergency rescue and cargo delivery. The system diagram is shown in Fig. 1.

1) Multiple Drones-Required Tasks: due to the limited capabilities of each UAV in low-altitude urban transaportation system, tasks are considered to be multiple drones-required tasks (MD tasks), which means that each task requires multi-UAVs, and each UAV can only perform one task.

2) Communication and Information Collection: the UAV a_i use the communication network to interact with neighboring UAVs, and it only know that assigned task and rewards.

B. Cooperative Mission Planning Model of Multiple UAVs

1) Kinematics: each UAV flies in a low-altitude urban environment when performing tasks, the kinematics model of the UAV a_i is as

$$\begin{cases} x_i(t+1) = x_i(t) + v_i(t)\Delta t \cos\psi_i(t) \sin\vartheta_i(t) \\ y_i(t+1) = y_i(t) + v_i(t)\Delta t \sin\psi_i(t) \sin\vartheta_i(t) \\ z_i(t+1) = z_i(t) + v_i(t)\Delta t \sin\vartheta_i(t) \\ v_i(t+1) = v_i(t) + u_i(t)\Delta t \\ \psi_i(t+1) = \psi_i(t) + u_{\psi_i}(t)\Delta t \\ \theta_i(t+1) = \theta_i(t) + u_{\theta_i}(t)\Delta t \\ \vartheta_i(t+1) = \vartheta_i(t) + u_{\vartheta_i}(t)\Delta t \end{cases} \tag{1}$$

where v_i is the velocity of the UAV a_i. (x_i, y_i, z_i) is the position of a_i. ψ_i, θ_i and $\vartheta_i(t)$ are the yaw angle, the roll angle and the pitch angle, respectively. u_i is the acceleration. Δt is the discrete time step. $u_{\psi_i}(t)$, $u_{\theta_i}(t)$ and $u_{\vartheta_i}(t)$ are the yaw angular velocity, the roll angular velocity, and the pitch angular velocity, respectively.

2) Task Rewards: A five-tuple formulation is introduced to analyze the cooperative task allocation and path planning of multi-UAVs within urban low-altitude airspace, where $\mathcal{A} = \{a_1, a_2, \ldots, a_n\}$ is the set of UAVs. $\mathcal{T} = \{T_1, T_2, \ldots, T_m\}$ is the set of tasks. $\mathcal{E} = \{E_1, E_2, \ldots, E_m\}$ is the central position of the task area. $\mathcal{O} = \{O_1, O_2, \ldots, O_{N_o}\}$ is the set of obstacles. $\mathcal{C} = \{C_1, C_2, \ldots, C_{N_c}\}$ is a set of constraints. Each UAV is equipped with a specific set of mission payloads to meet the operational requirements of diverse air traffic scenarios. The required load vector for executing the task T_j is $R^{T_j} = [R_1^{T_j}, R_2^{T_j}]$, where $R_1^{T_j}$ and $R_2^{T_j}$ are the number of type I and type II payloads required for task T_j, respectively. The task's payload demand is met until the total number of payloads carried by the UAVs is greater than or equal to the R^{T_j}. The payload carried by the a_i is given by

$$R^{a_i} = \left(R_1^{a_i}, R_2^{a_i}\right) \tag{2}$$

where $R_1^{a_i}$ and $R_2^{a_i}$ are the number of type I and type II payloads carried by the a_i, respectively.

Note that the initial value of the tasks are fixed value, the attack reward obtained when UAVs perform task T_j is defined as

$$B\left(T_j\right) = \sum_{i=1}^{\left|\mathcal{A}_{T_j}\right|} b_{ij} = \sum_{i=1}^{\left|\mathcal{A}_{T_j}\right|} d_{ij} r_{T_j} \tag{3}$$

where b_{ij} is the reward received by a UAV a_i from executing task T_j. $\mathcal{A}_{T_j}$ is the set of UAVs performing in the T_j. d_{ij} is the probability of successful delivery of task T_j by a_i, and r_{T_j} is the initial value of T_j.

3) Task Costs: each UAV carries a different payload and flies to the task area according to the corresponding path. Therefore, the task cost is composed of path cost and load cost. In the task allocation stage, we use Euclidean distance to estimate the initial path cost of each UAV. The initial path cost for the set of UAVs A_{T_j} is as

$$F\left(T_j\right) = \sum_{i=1}^{\left|\mathcal{A}_{T_j}\right|} f_i = \sum_{i=1}^{\left|\mathcal{A}_{T_j}\right|} \sqrt{\left(x_i - x_{T_j}\right)^2 + \left(y_i - y_{T_j}\right)^2 + z_i^2} \tag{4}$$

where f_i is the path cost of a_i. $\left(x_{T_j}, y_{T_j}\right)$ is the center coordinates of task T_j.

The load cost of UAVs is related to the payload carried and the time required to perform tasks [36]. The load cost of A_{T_j} for executing tasks T_j is given by

$$G(T_j) = \alpha_0 \left(1 - \beta_0^{\gamma_0 (t_{ji} + t_{\text{resp}})}\right) \sum_{i=1}^{\left|\mathcal{A}_{T_j}\right|} \frac{R^{T_j}}{R^{a_i}} \tag{5}$$

where α_0 and γ_0 are the load cost coefficients. β_0 is the load cost attenuation factor. t_{ji} is the time required for a_i to complete T_j, i.e., $t_{ji} = f_i / v_i$. t_{resp} is the response time.

4) Objective Function: according to Eqs. (2) to (5), a analysis of rewards and costs for cooperative multi-UAVs in low-altitude urban environments, the objective function for cooperative task allocation and path planning is as

$$\begin{aligned} \max J\left(T_j\right) &= \sum_{\forall T_j \in \mathcal{T}} B\left(T_j\right) - F\left(T_j\right) - G\left(T_j\right) \\ &= \sum_{a_i \in \mathcal{A}} \sum_{\forall T_j \in \mathcal{T}} U_i\left(T_j, \mathcal{A}_{T_j}\right) x_{ij} \end{aligned} \tag{6}$$

subject to

$$\begin{cases} \sum_{\forall T_j \in T} x_{ij} \le 1, \forall a_i \in \mathcal{A} \\ f_i \le f_{\max}, \quad v_{\min} \le v_i \le v_{\max} \\ \sum_{\forall T_j \in T} R^{T_j} \le \sum_{\forall a_i \in \mathcal{A}} R^{a_i} \\ R^{a_i} \in [1, 5], \quad u_{\psi_{\min}} \le u_{\psi_i} \le u_{\psi_{\max}} \\ r_{T_j} \in [15000, 30000], \quad u_{\theta_{\min}} \le u_{\theta_i} \le u_{\theta_{\max}} \\ d_{ij} \in [0.5, 0.8], \quad u_{\vartheta_{\min}} \le u_{\vartheta_i} \le u_{\vartheta_{\max}} \\ t_{\text{resp}} \in [0, 30], \quad z_i \le 200 \end{cases} \tag{7}$$

where $x_{ij} \in \{0, 1\}$ is a binary variable that mean whether the a_i will execute T_j. $f_{\max}$ is the maximum flight voyage of the UAV a_i. $v_{\min}$ and $v_{\max}$ are the minimum and the maximum velocity of a_i, respectively. $(u_{\psi_{\min}}, u_{\psi_{\max}})$, $(u_{\theta_{\min}}, u_{\theta_{\max}})$, and $(u_{\vartheta_{\min}}, u_{\vartheta_{\max}})$ are the minimum and maximum attitude angular velocity of a_i. $v_i \in [5, 10]$ m/s. $u_{\psi_i} \in [-1.5, 1.5]$ rad/s. $u_{\theta_i} \in [-1.2, 1.2]$ rad/s. $u_{\vartheta_i} \in [-1, 1]$ rad/s.

Noted that the UAV kinematics in Eq. (1) is used in the path planning stage. For task allocation, the optimization model is built based on distance and payload to to boost computational efficiency. As shown in Fig. 1, the path planning algorithm is adopted with additional practical physical constraints. This design compensates for the lack of explicit kinematic constraints in the task allocation stage.

C. Game Theory Model

In the game-theoretic formulation, UAVs are modeled as players with states characterized by their assigned tasks and relative positions in urban aerial scenarios. For computational tractability, UAVs are treated as point masses. The cooperative task allocation and path planning problem is a potential game $\Gamma = \{\mathcal{A}, \mathcal{S}, U_i\}$ and seek Nash equilibrium without introducing the nonlinear, continuous-time kinematics from Eq. (1) into the high-level decision-making process. This simplification does not imply that kinematic feasibility is ignored in the final solution. When an optimal solution is obtained, the low-level path planning module enforces the constraints of Eq. (1) and other physical limitations to generate executable and collision-free trajectories. This hierarchical decomposition ensures that plans are both globally coordinated and physically realizable.

The use of a point-mass model and Euclidean distance in the high-level task assignment does not imply neglecting the UAV dynamics, but rather represents a computationally tractable and conservative approximation. In low-altitude urban scenarios, the dynamic parameters of UAVs, such as velocity, acceleration, and turning radius—are bounded and relatively homogeneous. Under these conditions, the Euclidean distance preserves a monotonic relationship with feasible flight time and task cost, and can be used to evaluate task reachability and relative cost at the high-level stage.

Dynamic feasibility is further implicitly incorporated into the high-level model through conservative conditions derived from dynamic constraints, such as flight range, task reachability, and time-window constraints, as shown in Eq. (7), thereby avoiding obviously infeasible task assignments. The low-level path planning module strictly enforces the kinematic constraints, as well as physical limitations such as collision avoidance, to generate executable trajectories. If a task allocation is dynamically infeasible, it will be identified as infeasible during the low-level planning stage.

In this paper, the feasible sets of the high-level task assignment and the low-level path planning are consistent, and no non-executable task assignment results are observed. We focus on analyzing task allocation and path planning in the low-altitude urban traffic environments from the game theory perspective. Some theories is given which are necessary for the results.

1) Inhomogeneous Markov Chains: A Markov chain is inhomogeneous if the single-step matrix is not constant. The state transition probability matrix from time t to time k is as

$$H_{t,k} = \prod_{i=t}^{t+k-1} P_i \tag{8}$$

where the element in row a and column b is $h_{a,b}^{t,k}$.

Definition 1: An inhomogeneous Markov chain is weakly ergodic [14] if $k \to \infty$, for any t, a, a', b satisfies

$$\lim_{k\to\infty} \left| h_{a,b}^{t,k} - h_{a',b}^{t,k} \right| = 0 \tag{9}$$

Definition 2 (Scrambling matrix [37]): A state transition probability matrix P is a scrambling matrix, for any two rows of elements a and b, if there is a column of element γ, $p_{a\gamma} > 0$, $p_{b\gamma} > 0$ holds. The measure of the scrambling power is as

$$\text{sp}(P) = \min_{a,a'} \sum_b \min\left(p_{ab}, p_{a'b}\right) \tag{10}$$

Corollary 1: An inhomogeneous Markov chain, for each t, if the state transition probability matrix is regular [26], there must be an integer z_0 such that $H_{t,z_0} = \prod_{i=t}^{t+z_0-1} P_i$ is a scrambling matrix and $\text{sp}\left(H_{t,z_0}\right) > 0$.

Theorem 1 (Weakly ergodic [38]): A Markov chain is weakly ergodic, if and only if the time series can be divided into subsequences, as $k \to \infty$, it satisfies $\sum_{k=1}^{\infty} \text{sp}\left(H_{t_k,z_k}\right) > \infty$, where $z_k = t_{k+1} - t_k$.

Theorem 2 (Strongly ergodic [38]): A Markov chain P_t is strongly ergodic, if the following conditions hold: (a) Inhomogeneous Markov chains are weakly ergodic. (b) For each time t, P_t is a regular matrix with a steady state distribution $\boldsymbol{\pi}_t$. (c) $\boldsymbol{\pi}_t$ satisfies $\sum_{t=0}^{\infty} \sum_{s\in\mathcal{S}} |\pi_s(t) - \pi_s(t-1)| < \infty$.

2) Network Evolutionary Potential Game: We develop a network evolutionary game model $\Gamma = \{\mathcal{A}, \mathcal{S}, U_i\}$ to study collaborative mission planning, where player is $\mathcal{A} = \{a_1, a_2, \ldots a_n\}$, strategy is $\mathcal{S} = \{s_1, s_2, \ldots s_m\}$, which is consistent with $\mathcal{T}$. The utility function of a_i is U_i. Each UAV interacts with neighbors through a communication network, engages in group game within the entire cluster drone system, and obtains the utility based on the current group state at time t. Then, follow a strategy update rule to generate the group state at time $t+1$.

Definition 3 (Nash equilibrium [39]): A finite strategy game $\Gamma = \{\mathcal{A}, \mathcal{S}, U_i\}$ of n players with an action set $\mathcal{S}$ and utility function $\mathcal{U}$, a joint action $s^* = \left(s_1^*, s_1^*, \ldots, s_n^*\right)$ is Nash equilibrium if it holds that

$$U_i\left(s_i^*, s_{-i}^*\right) \geq U_i\left(s_i', s_{-i}^*\right) \tag{11}$$

where s_{-i}^* is the joint action of all individuals except for the individual a_i. s_i' is an action that are different from s^*.

Definition 4 (Potential game [40]): Γ is a potential game, if there are some potential functions ϕ, which satisfies

$$U_i\left(s_i, s_{-i}\right) - U_i\left(s_i', s_{-i}\right) = \phi\left(s_i, s_{-i}\right) - \phi\left(s_i', s_{-i}\right) \tag{12}$$

According to Definition 4, a potential game has at least one pure strategy Nash equilibrium s^*, i.e., $s^* = \arg\max_{s\in S} \phi(s)$. Each UAV $a_i \in A$ can obtain task information from neighbors, the global utility is defined as

$$U = \sum_{a_i\in\mathcal{A}} \sum_{\forall T_j\in T} U_i\left(T_j, \mathcal{A}_{T_j}\right) \tag{13}$$

Corollary 2: A finite strategy game Γ with a potential function $\phi(s)$ is a potential game that ϕ satisfies

$$\phi(s) = \sum_{a_i\in\mathcal{A}_{T_j}} U_i(s) \tag{14}$$

Proof: Let $U_{T_j}\left(s_i, s_{-i}\right)$ be the rewards obtained from the task T_j by $\mathcal{A}_{T_j}$. s_i^0 means that T_j was not assigned to a_i. We have

$$\begin{aligned} U_i\left(s_i, s_{-i}\right) &= U_{T_j}\left(s_i, s_{-i}\right) - U_{T_j}\left(s_i^0, s_{-i}\right) \\ &= \sum_{a_i\in\mathcal{A}_{T_j}} U_i\left(T_j, \mathcal{A}_{T_j}\right) - \sum_{a_k\in\mathcal{A}_{T_j}\backslash a_i} U_k\left(T_j, \mathcal{A}_{T_j}\backslash a_i\right) \end{aligned} \tag{15}$$

According to Eqs. (14) and (15), we have

$$
\begin{aligned}
&U_i(s_i, s_{-i}) - U_i(s_i', s_{-i}) \\
&= \sum_{a_i \in \mathcal{A}_{T_j}} U_i(T_j, \mathcal{A}_{T_j}) - \sum_{a_i \in \mathcal{A}'_{T_j}} U_i(T_j, \mathcal{A}'_{T_j}) \\
&= \phi(s_i, s_{-i}) - \phi(s_i', s_{-i})
\end{aligned} \tag{16}
$$

Therefore, according to Definition 1, the formulated game $\Gamma = \{\mathcal{A}, \mathcal{S}, U_i\}$ for task allocation of mult-UAVs leads to a potential game with $\phi(s)$ defined by Eq. (14).

Noted that s denotes a composite state consisting of the UAV a_i position and the information of the selected task, i.e., $s_i = (x_i, y_i, z_i, T_j)$. The optimal solution to the cooperative mission planning problem is a pure strategy Nash equilibrium in a finite strategy game $\Gamma = \{\mathcal{A}, \mathcal{S}, U_i\}$.

Corollary 3: The optimal solution to the cooperative task allocation and path planning problem is a pure strategy Nash equilibrium in a finite strategy game $\Gamma = (\mathcal{A}, \mathcal{S}, \mathcal{U})$.

Proof: We adopt proof by contradiction. Let s^* be the optimal solution to the mission planning problem. If is not the Nash equilibrium of Γ, there must be an s_i, which makes $U_i(s_i, s_{-i}^*) > U_i(s_i^*, s_{-i}^*)$. According to Corollary 2, Γ is a potential game, i.e., $\phi(s_i, s_{-i}^*) > \phi(s_i^*, s_{-i}^*)$. While $J(s_i^*, s_{-i}^*) = \sum_{j=1}^{m} \phi(s_i^*, s_{-i}^*) < \sum_{j=1}^{m} \phi(s_i, s_{-i}^*) = J(s_i, s_{-i}^*)$, which means that is not the optimal solution for task allocation and path planning in LU-ITS. However, this result contradicts the previous assumption, i.e., s^* is the Nash equilibrium.

IV. Methodology

A. Learning Algorithm in Potential Game

1) Log-Linear Learning Algorithm (LLA) [26]: We analyze the logit dynamics LLA. A potential game $\Gamma = \{\mathcal{A}, \mathcal{S}, U_i\}$, for each time t, UAV a_i randomly chooses a strategy $s_i \in \mathcal{S}$ with the same probability, which is defined as

$$
p_{s_i}(t) = \frac{\exp\{\beta U_i(s_i, s_{-i}(t-1))\}}{\sum\limits_{s' \in \mathcal{S}} \exp\{\beta U_i(s_i', s_{-i}(t-1))\}} \tag{17}
$$

where β is the Boltzmann parameter. As $\beta \to \infty$, for each t, a_i will select the task corresponding to the current maximum reward for updating, which is the best response learning algorithm (BRLA), i.e., $s_i^* \in \arg \sum\limits_{s_i \in S_i} \max U_i(s_i, s_{-i})$.

Given that a fixed $\beta \geq 0$, Γ has a homogeneous Markov chain with a state transition probability matrix is as

$$
P_t(s_i, s_i') = \frac{1}{n} \begin{cases} \sum_{i=1}^{n} p_{s_i'}(t), & s_i = s_i' \\ p_{s'}(t), & s_{-i} = s_{-i}' \text{ and } s_i \neq s' \\ 0, & \text{otherwise} \end{cases} \tag{18}
$$

Definition 5 (Steady state distribution [40]): A finite strategy game Γ with $\phi(s)$ if UAVs follow the Eq. (17), the Markov chain has a unique steady state distribution, that is

$$
\pi_s(t) = \frac{\exp\{\beta\phi(s)\}}{\sum\limits_{s' \in \mathcal{S}} \exp\{\beta\phi(s')\}} \tag{19}
$$

The steady-state distribution does not equate to a steady physical configuration of UAVs in the LUITS. Inversely, it is the equilibrium strategy distribution of the potential game. The game dynamics start from an initial strategy distribution $\pi_0(s)$ and evolve through a transient phase $\pi_t(s)$. This transient distribution captures the temporal adaptation of UAVs' strategies. Under finite potential game properties, $\pi_t(s)$ converges to the steady-state distribution $\pi^*(s)$.

Note that as $\beta \to \infty$, all weights on the steady state distribution $\pi_s(t)$ will be concentrated on the joint strategy corresponding to the maximum potential function $\phi(s)$, i.e., the optimal Nash equilibrium.

2) Improved Log-Linear Learning Algorithm (ILLA): To obtain the global optimal solution and ensure that the algorithm can converge with a probability one, a ILLA is proposed. For each time t, the selection of strategies for UAV a_i is as

$$
s_i(t) = \begin{cases} \text{random } s_i \in \mathcal{S}, \text{ with prob. } \varepsilon \\ \text{random } s_i \in \mathcal{S}_c, \text{ with prob. } p_{s_i}(t) \\ s_i(t-1), \text{ with prob. } 1 - \varepsilon - p_{s_i}(t) \end{cases} \tag{20}
$$

where ε is the exploration factor, $\mathcal{S}_c \subseteq \mathcal{S}$ is the strategy set of neighbors.

Additionally, an a time-independent Boltzmann parameter $\beta(t)$ is given by

$$
\beta(t) = \frac{\alpha \ln(t + \eta)}{c} \tag{21}
$$

Corollary 4: A networked evolutionary potential game $\Gamma = (\mathcal{A}, \mathcal{S}, \mathcal{U})$, if each UAV a_i adopts the logic dynamics of Eq. (17) and the strategy update mode of Eq. (20), where $\beta(t)$ follows Eq. (21), the inhomogeneous Markov chain induced by the ILLA is strongly ergodic.

Proof: Substituting Eq. (21) into Eq. (17), we have

$$
\begin{aligned}
p_{s_i}(t) &= \frac{(t+\eta)^{\frac{\alpha U_i(s_i, s_{-i}(t-1))}{c}}}{\sum\limits_{s_i' \in \mathcal{S}} (t+\eta)^{\frac{\alpha U_i(s_i', s_i(t-1))}{c}}} \\
&= \frac{1}{\sum\limits_{s_i' \in \mathcal{S}} (t+\eta)^{\frac{\alpha\{U_i(s_i', s_i(t-1)) - U_i(s_i, s_i(t-1))\}}{c}}}
\end{aligned} \tag{22}
$$

Note that $0 < \alpha < 1$, $\eta > 0$, $c > 0$, for any $t \geq 1$, $t + \eta > 1$ holds. Let $|U_i(s_i', s_{-i}(t-1)) - U_i(s_i, s_{-i}(t-1))| = M_i$, where M_i is a bounded constant. By adjusting the parameters α, for any $s_i, s_{-i} \in \mathcal{S}$, $\alpha M_i \leq 1$ holds. Therefore, we have

$$
p_{s_i}(t) = \frac{1}{\sum\limits_{s_i \in \mathcal{S}} (t+\eta)^{\frac{\alpha M_i}{c}}} \geq \frac{1}{n(t+\eta)^{1/c}} \tag{23}
$$

According to Definition 5, Eqs. (20) and (23), we have

$$
\begin{aligned}
P_t(s_i, s_i') = &\prod_{s_i(t) \in \mathcal{S}} \frac{\varepsilon}{|\mathcal{S}|} \prod_{s_i(t) = s_i(t-1)} \frac{1}{\sum\limits_{s_i' \in \mathcal{S}} (t+\eta)^{\frac{\alpha M_i}{c}}} \\
&\times \prod_{s_i(t) \in \mathcal{S}_c} \frac{1}{\sum\limits_{s_i' \in \mathcal{S}_c} (t+\eta)^{\frac{\alpha M_i}{c}}} \geq \frac{\varepsilon}{|\mathcal{S}| n_c (n - n_c)(t+\eta)^{1/c}}
\end{aligned} \tag{24}
$$

where $|\mathcal{S}|$ is the maximum cardinality of the $\mathcal{S}$. n_c is the number of neighbors of a_i.

According to Definition 1 and Corollary 1, the positive elements in the state transition probability matrix of step c starting from time t is given by

$$h_{s_1,s_2}^{*(t,c)} = \sum_{b_1\in\mathcal{S}} \cdots \sum_{b_c\in\mathcal{S}} P_t(s_1,b_1)\cdots P_t(b_c,s_2) \geq \frac{\varepsilon^c}{(|\mathcal{S}|n_c(n-n_c))^c(t+\eta+c-1)} \tag{25}$$

According to Definition 2 and Eq. (10), we have

$$\mathrm{sp}\left(H_{t,c}\right) \geq \frac{\varepsilon^c}{(|\mathcal{S}|n_c(n-n_c))^c(t+\eta+c-1)} \tag{26}$$

Let $t = (k-1)c + 1$,

$$\sum_{k=1}^{\infty} \mathrm{sp}\left(H_{t,c}\right) \geq \left(\frac{\varepsilon}{|\mathcal{S}|n_c(n-n_c)}\right)^c \sum_{k=1}^{\infty} \frac{1}{ck+\eta} = \infty \tag{27}$$

According to Theorem 1, the inhomogeneous Markov chain defined by the ILLA is weakly ergodic. From the above analysis and Eq. (18), the $\pi_s(t)$ is as

$$\pi_s(t) = \frac{(t+\eta)^{\frac{\alpha\phi(s)}{c}}}{\sum_{s'\in\mathcal{S}}(t+\eta)^{\frac{\alpha\phi(s')}{c}}} \tag{28}$$

$$\sum_{t=0}^{\infty}\sum_{s\in\mathcal{S}}|\pi_s(t+1)-\pi_s(t)| < \infty \tag{29}$$

Algorithm 1 Improved Log-Linear Learinning Algorithm

Input: n of UAVs, m of tasks, Coordinate (x_i, y_i) of a_i, Coordinate (x_{T_j}, y_{T_j}), $t_{\max}$, strategy set $\mathcal{S}$.
Output: The UAV set $\mathcal{A}_{T_j}$ for each task. U and U_i.
Initialization: m of tasks are assigned to n of UAVs, randomly. $\varepsilon = \mathrm{rand}(0, 0.1)$. $x_{ij} = 0$. $c = \mathrm{rand}(0, t_{\max})$. $\alpha = \mathrm{rand}\left(1\times10^{-4}, 1\times10^{-3}\right)$. $\eta = \mathrm{rand}(1, 10)$.
While $t \leq t_{\max}$ **do**
  for $a_i \in \mathcal{A}$ **do**
    if $x_{ij} = 0$ **then**
      Calculate $s_i(t)$ using Eq. (20);
      $t \leftarrow t + 1$;
    else
      Calculate $\beta(t)$ using Eq. (21);
      Broadcast the local information to neighbors;
      Calculate U and U_i using Eqs. (13) and (15);
    end if
    Update s_i and $U_i(s_i, s_{-i})$;
    if $U_i(s_i, s_{-i}) > U_i\left(s_i', s_{-i}\right)$ **then**
      Find Nash equilibrium $s_i^* \leftarrow s_i$, obtain $\mathcal{A}_{T_j}$, U_i;
    else
      Adjust the $\beta(t)$ to generate new α and η;
    end if
  end for
end while
Return $\mathcal{A}_{T_j}$ for each task. U and U_i.

Therefore, according to Theorem 2, the Markov chain is strongly ergodic. From Corollary 4, the following theorem describes the results of this section, and the pseudocode of ILLA is shown in Algorithm 1.

Theorem 3: A networked evolutionary potential game Γ, for the proposed ILLA with $\beta(t)$, if the strategy of Eq. (20) is adopted, the ILLA can converge to the solution set corresponding to the maximum potential function with a probability one, i.e., the optimal Nash equilibrium.

$$\lim_{t\to\infty} \mathrm{Prob.}\left\{s(t) \in \left\{s^* \mid \phi(s^*) = \max_{s\in\mathcal{S}}\phi(s)\right\}\right\} = 1 \tag{30}$$

B. Path Planning Algorithm

Path planning is another problem in cooperative task planning for multi-UAVs. In this section, based on the A-Star algorithm, we propose a Constraint-Based Multi-layer Bidirectional Adaptive A-Star (CBMBA A-Star) algorithm to find a collision-free path for each UAV.

1) Conventional A-Star Algorithm: the A-Star algorithm is a deterministic graph-based search algorithm that can ultimately obtain the optimal solution of the path. The heuristic function of the conventional A-Star algorithm is given by

$$f(k) = g(k) + h(k) \tag{31}$$

where $f(k)$ is total cost. $g(k)$ is actual cost from the start to the node k. $h(k)$ is estimated cost from node k to the goal.

2) Constraint-Based Multi-Layer Bidirectional A-Star (CBMBA A-Star): the conventional A-Star algorithm uses a single expansion method to calculate the straight-line distance between the current node and neighboring nodes, which can result in the generation of inaccurate and unsafe paths. For the omnidirectional search, the computational complexity of the algorithm is significantly increased when the planning space is large. And the path rarely meets the real flight constraints. Hence, We propose the CBMBA A-Star algorithm with a new heuristic function and search strategy. The pseudocode of CBMBA A-Star algorithm is shown in Algorithm 2.

a) The improved heuristic function:

$$f(k) = g(k) + \left(1 + \frac{g(k)}{g(k)+h(k)}\right)h(k) \tag{32}$$

The improved heuristic function aims to enhance the accuracy of the solution by refining the weight of the heuristic term $h(k)$. By increasing its contribution, the more accurate estimations of the remaining cost are prioritized, which can lead to better performance in terms of finding higher-quality solutions.

Noted that although increasing the weight of $h(k)$ may affect the consistency of the heuristic function in the A* algorithm, as long as the improved heuristic remains admissible, the global optimality of A* is still guaranteed. Specifically, in Eq. (32), according to the proposed improvement, the weight of $h(k)$ varies dynamically within a relatively small range. In many cases, the heuristic remains admissible and never overestimates the actual cost. Therefore, in certain applications, allowing a slight compromise in consistency can enhance the solution quality, achieving a balance between solution accuracy and algorithmic consistency.

b) Search strategy: to improve the search efficiency, adaptability to the environment, and path quality, considering the flight constraints of UAVs turning and the characteristics

Algorithm 2 Constraint-Based Multi-Layer Bidirectional Adaptive A-Star Algorihm

Input: $\mathcal{A}_{T_j}$, (x_i, y_i), (x_{T_j}, y_{T_j}).
Output: $\mathbf{P}_i$ (Path sequence of each UAV a_i), $f(k)$.
Initialization: Generate two OPEN_lists and CLOSE_lists. (x_i, y_i) and (x_{T_j}, y_{T_j}) are added to two OPEN_lists.
While $k \le k_{\max}$ **do**
  for $\mathcal{A} = \{a_1, a_2, \ldots, a_n\}$ **do**
    A multi-layer bidirectional search using Eqs. (32) and (33);
    $k \leftarrow k + 1$;
    Generate nodes of virtual circle based on (x_k, y_k);
    if *collision detected* **do**
      Obtain node (x_{k+1}, y_{k+1}) using $\lambda_{\min}$;
      OPEN_list $\leftarrow \left(x^*_{k+1}, y^*_{k+1}\right) \in \arg\min f(k+1)$;
    else
      Track search with adaptive λ using Eq. (34);
    end if
    if *the current node in OPEN_lists is same* **then**
      Track their parent nodes and obtain $\mathbf{P}_i$;
    else
      Traverse all adjacent nodes of (x_k, y_k);
      OPEN_list $\leftarrow \left(x^*_k, y^*_k\right) \in \arg\min f(k+1)$;
      Update $\mathbf{P}_i$, $f(k)$;
    end if
  end for
end while
Return $\mathbf{P}_i$, $f(k)$.

of bandit threats, we propose a constraint-based multi-layer extension and bidirectional adaptive step search strategy.

A sector area is adopted instead of the omnidirectional search to meet physical constraints. Meanwhile, dividing the sector area into multiple small pieces can effectively avoidance the static radar threat when the map's grid is large. Then, each UAV adopts a bidirectional search strategy with two OPEN and CLOSE lists, which further improves the algorithm search efficiency. Furthermore, a small virtual circle is constructed with the current node as the center, which determines the safety of the current node during each search. The search step λ is dynamically adjusted based on the position of the corresponding path node to facilitate the handle bandit threats and improve the real-time performance of the algorithm. The coordinates of the path nodes is given by

$$\begin{cases} x_{k+1} = x_k + L_i \cos\rho_i \\ y_{k+1} = y_k + L_i \sin\rho_i \end{cases} \tag{33}$$

where ρ_i and L_i are the angles and arc lengths between the search line segments, respectively.

$$\lambda = \begin{cases} \lambda_{\min}, & r_i(k) \le r \\ \dfrac{\lambda_{\max}}{1 + r/r_i(k)}, & 5\ r < r_i(k) < r \\ \lambda_{\max}, & r_i(k) \ge 5\ r \end{cases} \tag{34}$$

where r is the radius of a virtual small circle, $r_i(k)$ is the distance from the k th path node of UAV a_i to the equivalent surface of the threat. $\lambda_{\max}$ and $\lambda_{\min}$ are the maximum and minimum search steps, respectively.

V. Simulation Results and Discussion

A. Mission Scenarios

To address the problem of multi-UAVs collaborative task allocation and path planning in low-altitude urban air traffic environments, two typical mission scenarios for simulation analysis, i.e., low-altitude urban air emergency rescue (Case I) and last-mile cargo transportation and delivery (Case II). UAVs and task areas are randomly deployed across the simulated low-altitude urban area, with the area dimensions set to 600m×600m and 1000m×1000m for the two cases respectively. The environment incorporates typical urban constraints including building obstacles and temporary no-fly zones to further validates the model and algorithm.

B. Case I: Low-Altitude Urban Air Emergency Rescue Task

20 UAVs are considered to execute three tasks in low-altitude urban air traffic environments, which aims to simulate emergency responses to urban sudden traffic incidents. The task contains the accident scene monitoring (T_1), emergency medical supplies delivery (T_2), traffic flow reporting (T_3). Simulation experiments are conducted to verify the effectiveness of the proposed the ILLA method and the CBMBA A-Star algorithm, where the LLA [26], the BRLA [25], and the CBBA [17] methods are used as comparisons in the task allocation. While the A-Star algorithm and differential evolution (DE) algorithm [19] are adopted as comparisons in path planning.

Fig. 2 illustrates the cooperative task execution of 20 UAVs in a low-altitude urban traffic environment, performing three emergency traffic rescue tasks, where small circles represent individual UAVs, with connecting lines indicating the communication links among them. Hexagonal stars denote the locations of the tasks, and UAVs sharing the same color are assigned to the corresponding task. Fig. 2a depicts the initial stage, where tasks are randomly assigned to a subset of UAVs. Fig. 2b shows the task allocation after 67 iterations, demonstrating that all UAVs have been successfully coordinated to complete the tasks efficiently.

Fig. 3 illustrates the global utility of task allocation for 20 UAVs collaboratively executing three traffic emergency rescue tasks in a low-altitude urban traffic environment. Compared to other algorithms, the ILLA algorithm ensures the maximum global utility in task allocation. Within a limited time, both the LLA and the BRLA algorithms converge to their respective values, with their global utility curves exhibiting a non-decreasing characteristic. Although the BRLA algorithm converges more quickly, it results in lower utility.

Table II illustrates the differences in task rewards and execution time across various algorithms, with the CBBA method serving as a benchmark for comparison. It is clear that both CBBA and LLA exhibit relatively low task execution efficiency in the context of emergency rescue operations in low-altitude urban traffic environments. Inversely, the ILLA algorithm outperforms others in both execution time and task

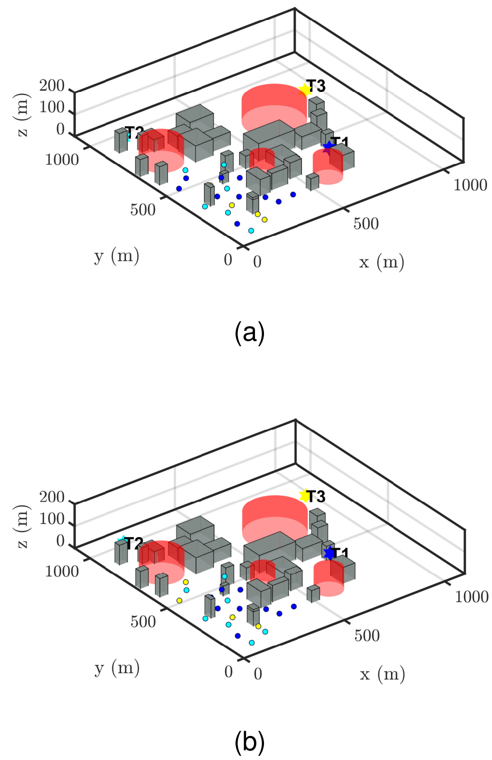

Fig. 2. Task allocation by using the ILLA in Case I. (a) t = 1. (b) t = 67.

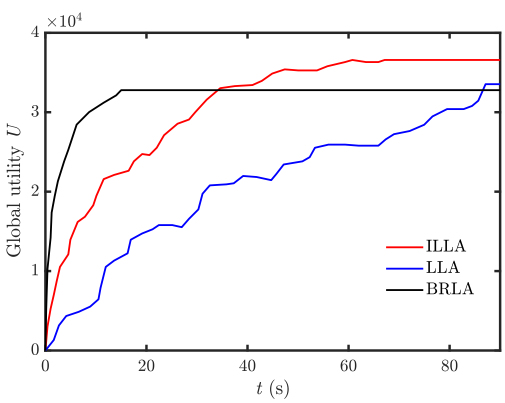

Fig. 3. The global utility of task allocation.

rewards, further validating the superiority of our game theory-based framework and the proposed algorithm.

Fig. 4 shows that the CBMBA A-Star algorithm successfully find paths for each UAV, avoiding building obstacles and no-fly zones in low-altitude urban traffic during emergency rescue operations. However, other two algorithms rarely achieve safe path planning for all UAVs. Table III highlights that both the bidirectional sector A-Star and the differential evolution algorithms achieve lower path costs compared to the A-Star algorithm. In terms of computation time, the CBMBA A-Star algorithm stands out with significant advantages. Despite the multi-layer sector search strategy increasing the number of path nodes, it results in fewer turning points, meaning smoother paths that better meet flight requirements.

TABLE I

NOMENCLATURE

Symbols	Significance
v_i	The velocity of UAV a_i.
s_i	The state of a_i (Location and task sequences).
U_i	The utility of a_i (reward).
s_{-i}	Joint states of all drones except for the a_i. (Location and task sequences of other UAVs)
s_i^*	Nash equilibrium strategy of a_i. (A stable state: corresponding position and task)
$H_{t,k}^*$	The state transition probability (between two states).
s_i'	The different state (strategy) from s_i. (a_i chooses other tasks and positions)
$\mathcal{S}$	The set of all states. (All available locations and tasks to choose from).
$\mathcal{A}_{T_j}$	The set of UAVs that peforme to task T_j.
$\mathcal{T}$	The set of tasks.
r_{T_j}	The value of the task T_j.
m	The number of tasks.
n	The number of UAVs.
R^{T_j}, R^{a_i}	The payload demand, the payload carried by a_i.

TABLE II

PERFORMANCE OF TASK ALLOCATION ALGORITHMS IN CASE I

Indicators	ILLA	LLA	BRLA	CBBA
Task rewards	36578.95	33531.78	32778.94	32641.28
Execution time (s)	66.91	88.74	14.86	112.38

TABLE III

PERFORMANCE OF PATH PLANNING ALGORITHMS IN CASE I

Algorithm	Path cost (m)	Run time (s)	Path node	Turning node
CBMBA A-Star	14529.28	21.43	1208	42
A-Star	14907.63	50.31	1578	163
DE	14696.83	67.25	695	51

This further demonstrates the effectiveness of the CBMBA A-Star algorithm in the traffic rescue scenarios.

C. Case II: Last-Mile Cargo Transportation and Delivery Task

40 UAVs are considered to perform five tasks in the low-altitude urban air traffic scenario with some urban buildings, no fly zones, and two unexpected no-fly zones, where five task areas are express delivery stations in different communities and commercial buildings. To test the performance of the ILLA and the CBMBA A-Star algorithms, the LLA, the BRLA, and the CBBA method are adopted as comparison algorithms in the task allocation. However, the dynamic A-Star (D-Star) [41] and the MSFDE algorithm [19] are applied as comparisons in the path planning.

From Fig. 5 and Table IV, as the size of the UAVs increases, compared to the other methods, the task rewards obtained by the ILLA improve 13.57%, 15.64%, and 11.35%, respectively. And compared with the LLA and the CBBA, the task execution time decreased by 50.34% and 76.31%, respectively. Therefore, this further validates the ILLA method's optimality,

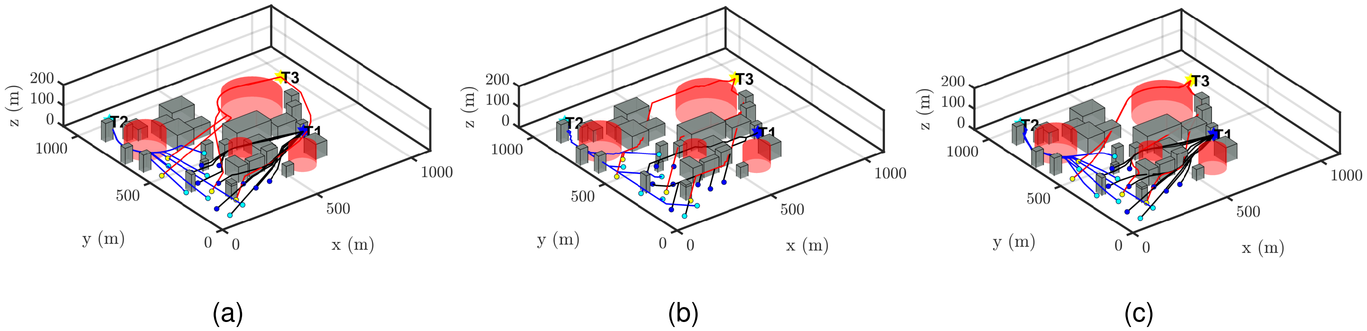

Fig. 4. Cooperative path planning of multi-UAVs in Case I. (a) CBMBA A-Star algorithm. (b) A-Star algorithm. (c) DE algorithm.

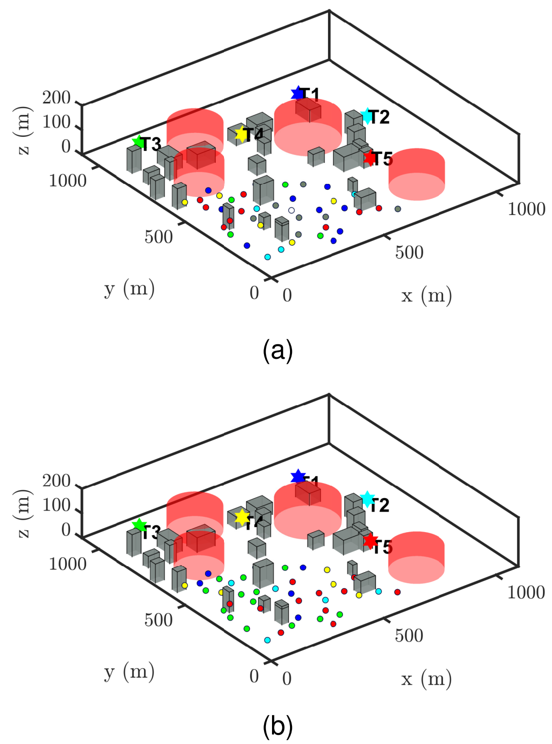

Fig. 5. Task allocation by using ILLA in Case II. (a) t = 1. (b) t = 150.

TABLE IV
PERFORMANCE OF TASK ALLOCATION ALGORITHMS IN CASE II

Indicators	ILLA	LLA	BRLA	CBBA
Task rewards	69862.14	60226.08	58781.29	61772.4
Execution time (s)	149.31	224.95	37.62	263.25

good computational efficiency, and scalability in the low-altitude urban traffic scenarios.

Fig. 6 presents the original path planning and re-planning results using different algorithms in the last-mile cargo delivery and transportation under the threats of unknown areas, represented as blue hemispheres. By using the CBMBA A-Star method some UAVs adjust their paths to avoid these architectural barriers and no fly zones, with the dashed lines indicating the online re-planning paths. Notably, other algorithms cannot obtain a safe path. The CBMBA A-Star algorithm results in the fewest UAVs needing to re-plan their paths, further demonstrating the effectiveness of the proposed CBMBA A-Star algorithm in the low-altitude urban environments for cargo transportation and delivery.

TABLE V
PERFORMANCE OF PATH PLANNING ALGORITHMS IN CASE II

Situations	Algorithm	Path cost (m)	Run time (s)	Turning node
Original planning	CBMBA A-Star	27154.01	67.47	57
	A-Star	28952.46	160.56	169
	DE	27651.29	129.17	74
Path re-planning	CBMBA A-Star	27815.92	89.28	79
	D-Star	29872.67	347.85	211
	MSFDE	28370.59	206.51	106

Table V shows that the path cost obtained by the CBMBA A-Star algorithm increased by only 2.43% during the path re-planning stage, representing the smallest increase among the three algorithms. Furthermore, the CBMBA A-Star algorithm also shows significant advantages in terms of running time and path smoothness. Compared with the original planning, the number of turning nodes of the CBMBA A-Star algorithm increased by only 22 during the path re-planning stage. Inversely, the D-Star algorithm resulted in increases of 7.39% in path cost, 167.08% in running time, and 74.35% in the number of turning nodes, respectively. Similarly, the MSFDE algorithm exhibited increases of 1.99% in path cost, 56.79% in running time, and 34.18% in the number of turning nodes, respectively. From the above statistical results, it is evident that the proposed CBMBA A-Star algorithm achieves the optimal performance in path r-eplanning. Therefore, local online replanning by using the CBMBA A-Star algorithm can yield a safe path, validating the optimality and real-time performance of our method.

Fig. 7 presents the statistical results of average iteration times as the number of UAVs and tasks increases in the low-altitude urban last-mile cargo transportation and delivery scenario, with the simulations conducted using Monte Carlo for 100 instances. From Figure 7a, as the number of UAVs grows, each UAV requires more information exchange and computation to achieve globally optimal collaborative task

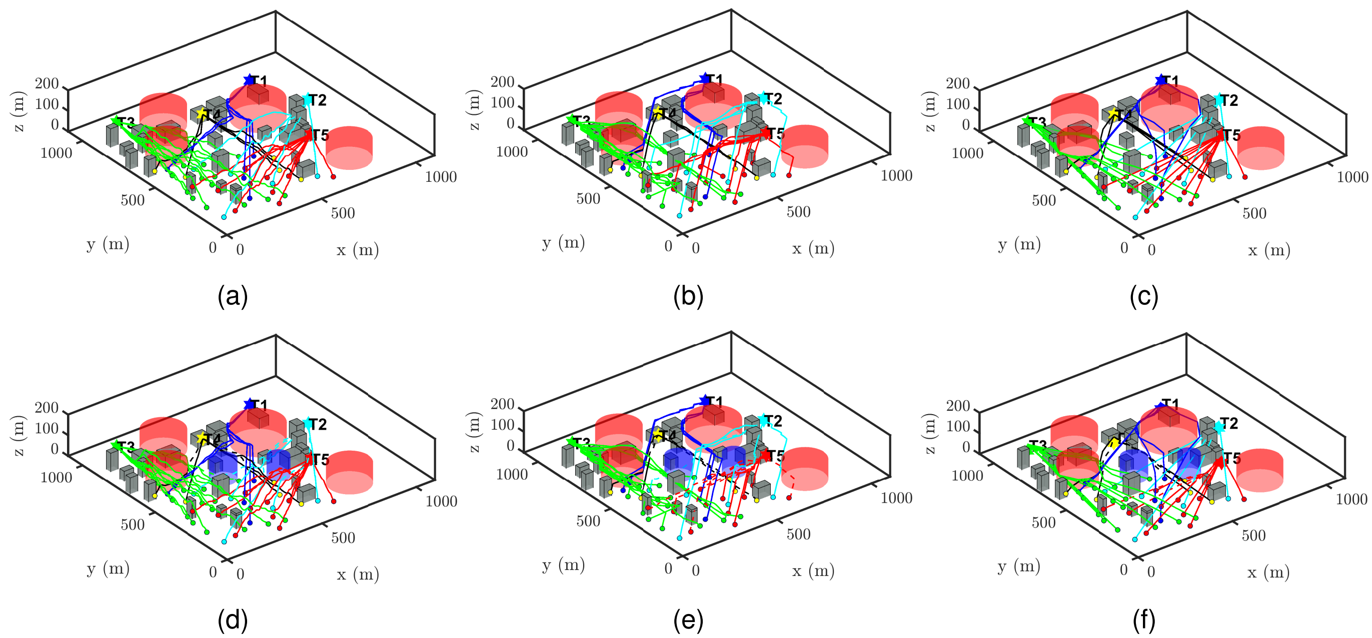

Fig. 6. Cooperative path planning results of multi-UAVs in Case II. (a)~(c): Original path planning. (b)~(f): Path re-planning. (a) CBMBA A-Star algorithm (Original planning). (b) A-Star algorithm. (c) DE algorithm. (d) CBMBA A-Star algorithm (Re-planning). (e) D-Star algorithm. (f) MSFDE algorithm.

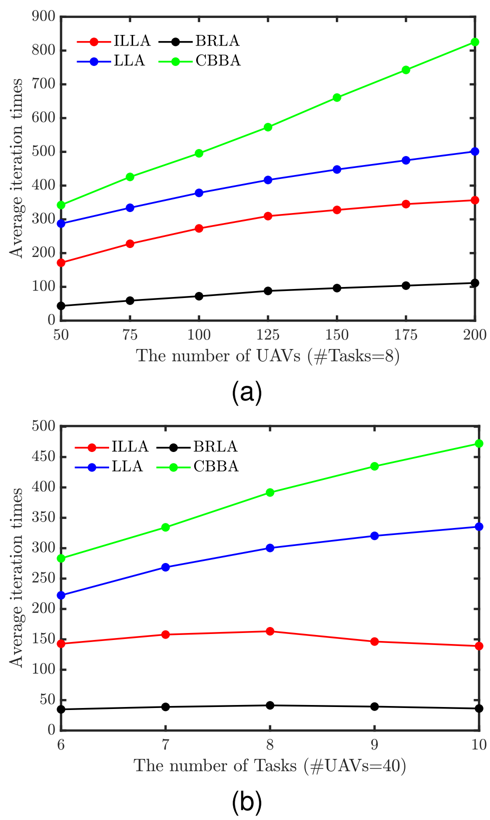

Fig. 7. (a) The total number of UAVs. (b) The total number of tasks versus average iteration times.

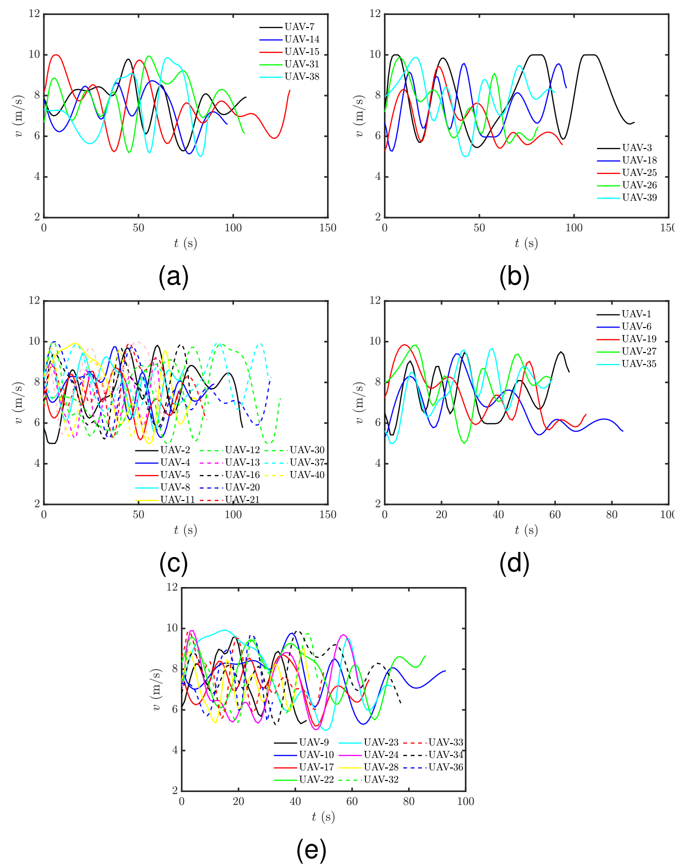

Fig. 8. The velocity of 40 UAVs in Case II. (a) Task T_1. (b) Task T_2. (c) Task T_3. (d) Task T_4. (e) Task T_5.

allocation and path planning. This highlights the significant impact on task allocation and path planning efficiency in low-altitude urban traffic environments, particularly for last-mile cargo transportation and delivery. Figure 7b indicates that the average iteration times of the ILLA algorithm decrease when the number of tasks exceeds eight. The likely reason is that an increase in task numbers provides more options for the UAVs, which helps mitigate conflicts between task allocation and path planning. This further demonstrates the scalability and robustness of the proposed model and algorithm, confirming their effectiveness in low-altitude urban traffic environments.

To further verify the physical feasibility of each UAV's trajectory, an analysis of the discrete-time variations in the UAV's velocity v_i are conducted, as depicted in Fig. 8.

In Fig. 8, all UAVs involved in the tasks maintain their velocities stably within the predefined interval $[v_{\min}, v_{\max}]$ with relatively smooth variation trends and no abrupt fluctuations that would lead to excessive speed. Such smooth velocity changes indicate that the UAVs do not need to perform drastic attitude adjustments during the cargo delivery process, a critical advantage for last-mile delivery, as it helps prevent cargo damage caused by sudden attitude changes and ensures the stability of the delivery process.

VI. Conclusion

This paper presents a distributed autonomous decision-making framework for multi-UAV cooperative task allocation and path planning in complex low-altitude urban environments. To achieve optimal task allocation and ensure system stability, a network evolutionary potential game model is established to link task allocation with game learning. Additionally, an ILLA algorithm with a new update strategy and a time-independent Boltzmann parameter is proposed, enabling the algorithm to find the Nash equilibrium with probability one. Furthermore, a CBMBA A-Star algorithm is introduced to provide optimal and collision-free paths for UAVs. Simulation results demonstrate that the ILLA algorithm minimizes task execution time while maximizing global task rewards, and the CBMBA A-Star algorithm reduces path cost and computation time. As the number of UAVs and tasks increases, the ILLA algorithm effectively mitigates UAV conflicts, while the CBMBA A-Star algorithm adaptively adjusts the search step to enable online path replanning in dynamic scenarios, especially in environments with unknown threat areas. The numerical results validate the effectiveness and superiority of the proposed model and algorithms in traffic emergency rescue and last-mile delivery tasks in low-altitude urban environments.

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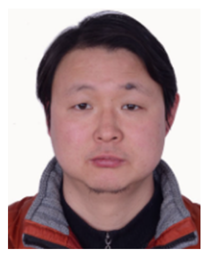

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